SONiX 32-Bit Cortex-M0 Micro-Controller

DMA CT16B Example Code

AMENDENT HISTORY

Version	Date	Description
1.0	2024/11/27	1. First version.

Table of Content

1 OVERVIEW

This document provides our customers the C program examples of CT16B in DMA mode.

2 FEATURE

- CT16B PWM function with DMA mode example.
- Use ARM-compiler V6.12

3 HW

- The starter kit board of vendor desired MCU.

4 EXAMPLE CODE

4.1 Initialization

The CT16B0 and DMA functions are initialized in the main code. The example is to use DMA to transfer SRAM data to CT16B MR0~3. Each time TC is matched with MR9, PWM0~3 will change the duty cycle.

```
//----------------------------------------------------------------------
//User Code starts HERE!!!

// Init CT16B0 & DMA
DMACT16B0PWM0123Init();
```

The setting of DMA should refer to the table in the specifications.

```
void DMACT16B0PWM0123Init(void)
{
  DMA_InitSt stDMACH_Init;

  // Set DMA CH0 as M(SRAM) to P(CT16B0 PWM0~3) mode
  // Source : Memory, Increment, 16bit, no request, Burst 4
  stDMACH_Init.b_SrcAddrCtrl = DMA_SRCAD_CTL_INC;
  stDMACH_Init.b_SrcMode     = DMA_SRC_MEMORY;
  stDMACH_Init.b_SrcWidth    = DMA_SRC_WIDTH_16BIT;
  stDMACH_Init.b_SrcReqSel   = DMA_SRCRS_NONE;
  stDMACH_Init.b_BurstSize   = DMA_BURST_4;
  // Destination : Peripheral CT16B0, Increment cyclic, 16bit, CT16B0 PWM0~3
  stDMACH_Init.b_DstAddrCtrl = DMA_DSTAD_CTL_CYC;
  stDMACH_Init.b_DstMode     = DMA_DST_PERIPHERAL;
  stDMACH_Init.b_DstWidth    = DMA_DST_WIDTH_16BIT;
  stDMACH_Init.b_DstReqSel   = DMA_DSTRS_CT16B0_MR0;
  // Channel : LV0, Enable TC/ABT interrupt
  stDMACH_Init.b_Priority    = DMA_CHPRI_LV0;
  stDMACH_Init.b_IntTCEn     = DMA_INT_TC_MSK_DIS;
  stDMACH_Init.b_IntABTEn    = DMA_INT_ABT_MSK_DIS;

  // Initial DMA
  DMA_Init(&stDMACH_Init, eDMA_CH0);
```

The PWM period is set to 1ms, and the initial duty cycles of PWM0~3 are 10%, 30%, 70% and 90% respectively.

```
    // Initial CT16B
    CT16B0_Init();

    //Set MR9 value for 1ms period ==> count value = 1000*12 = 12000
    SN_CT16B0->MR9 = 12000;

    //Set MR0/1/2/3 value as 10%/30%/70%/90% duty
    SN_CT16B0->MR0 =  1200;
    SN_CT16B0->MR1 =  3600;
    SN_CT16B0->MR2 =  8400;
    SN_CT16B0->MR3 = 10800;

    //Enable PWM function, IOs and select the PWM modes
    SN_CT16B0->PWMCTRL =
      (mskCT16_PWM0EN_EN | mskCT16_PWM0MODE_2 | mskCT16_PWM0IOEN_EN)| //Enable PWM0 function, IO and select as PWM mode 1
      (mskCT16_PWM1EN_EN | mskCT16_PWM1MODE_2 | mskCT16_PWM1IOEN_EN)| //Enable PWM1 function, IO and select as PWM mode 1
      (mskCT16_PWM2EN_EN | mskCT16_PWM2MODE_2 | mskCT16_PWM2IOEN_EN)| //Enable PWM2 function, IO and select as PWM mode 1
      (mskCT16_PWM3EN_EN | mskCT16_PWM3MODE_2 | mskCT16_PWM3IOEN_EN); //Enable PWM3 function, IO and select as PWM mode 1

    //Set MR9 match TC reset
    SN_CT16B0->MCTRL = mskCT16_MR9RST_EN;

    //MR0 DMA request can issue.
    SN_CT16B0->DMA = mskCT16_MR0_DMA_EN;

    //Set CT16B3 as the up-counting mode.
    SN_CT16B0->TMRCTRL = (mskCT16_CRST | mskCT16_CM_EDGE_UP);

    //Wait until timer reset done.
    while (SN_CT16B0->TMRCTRL & mskCT16_CRST);
}
```

4.2 DMA CT16B function enable

DMA request will be issued when the TC matches MR0, and the new MRn value will be loaded when TC=0. The DMA TC flag and interrupt will be set when all data transfers are completed.

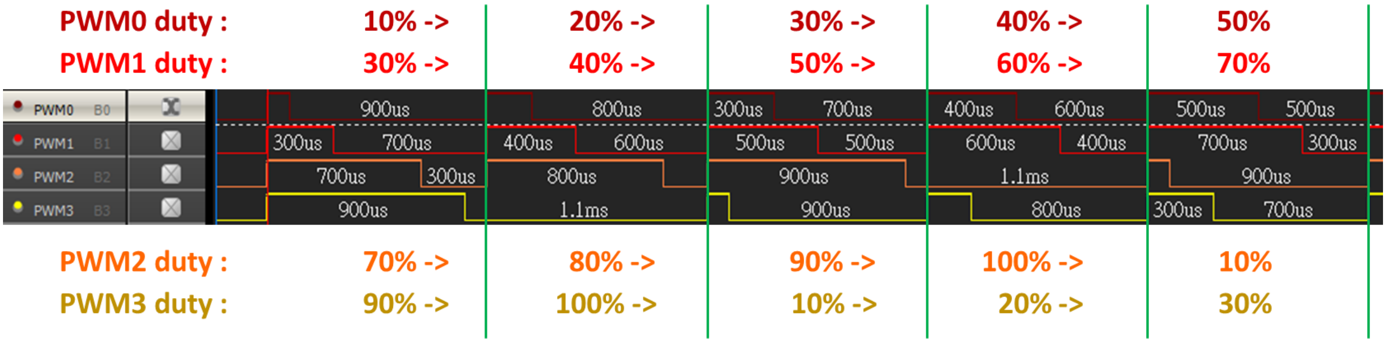
```
    // Start DMA CT16B0 PWM0~3 transfer
    DMACT16B0PWM0123Start();

    // Wait for DMA TC interrupt flag
    while(stDMA0IntFlag[eDMA_CH0].Flag.bits.TC != 1);

    while(1); // Pass
}
```

5 RESULT

Each time the PWM cycle is completed, the new duty value will be updated. According to the data in SRAM, the PWM duty will increase by 10% after DMA transfer The results can be observed through PWM0~PWM3 Pins.



Main Office:
Address: 10F-1, NO. 36, Taiyuan Street, Chupei City, Hsinchu, Taiwan R.O.C.
Tel: 886-3-5600 888
Fax: 886-3-5600 889

Taipei Office:
Address: 15F-2, NO. 171, Song Ted Road, Taipei, Taiwan R.O.C.
Tel: 886-2-2759 1980
Fax: 886-2-2759 8180

Hong Kong Office:
Unit No.705, Level 7 Tower 1,Grand Central Plaza 138 Shatin Rural Committee Road, Shatin, New Territories, Hong Kong.
Tel: 852-2723-8086
Fax: 852-2723-9179

Technical Support by Email:
Sn8fae@sonix.com.tw